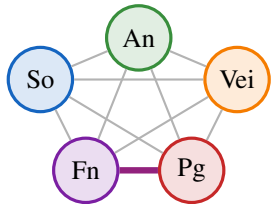


## Living material



variational multiphase model  
15 unknown interactions  $a_{ij}$

## Physics-constrained Bayesian inference

composition

+ viability

→  
TMCMC on a GPU (JAX,  $\sim 150\times$ )  
3 seeds per stage, explicit  
convergence & identifiability checks  
→

## Identified interactions

[posterior of A /  
identifiability /  
knockout prediction  
after final runs]